

## GE261004 DESIGN THINKING

Unit IV – Prototyping and Testing · Presentation 5 of 6

# Testing Methods and User Feedback

What a test is for · thirty minutes with one user · how many users you need · what to say and what not to · capturing feedback a team can act on

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# In this presentation

- What testing is for
- Thirty minutes with one user
- How many users you need
- Writing the task
- What to say, what not to say
- The feedback capture grid
- Filling the grid
- A/B comparison

# 1

## Running a test

Watching somebody use it, and saying almost nothing

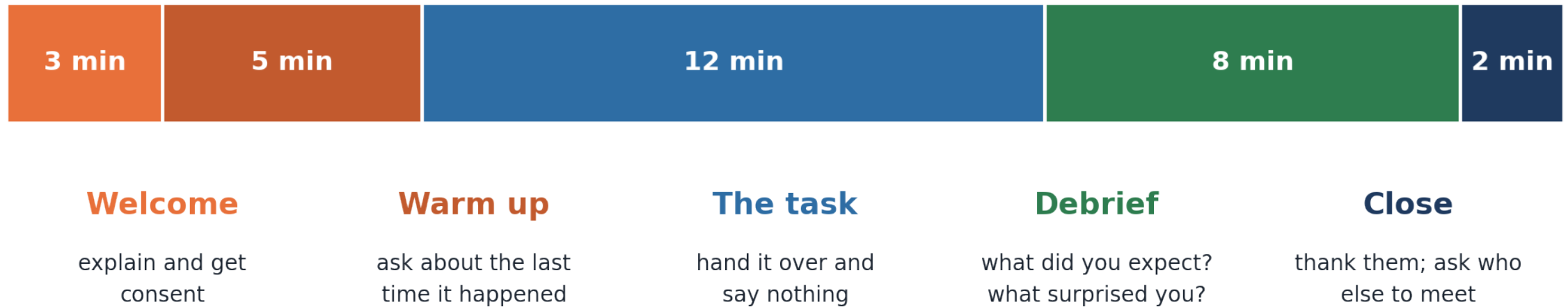
# What testing is for

- To find out what people do, not whether they like it
- Liking is easy to get and means nothing; behaviour is hard to fake
- You are testing the prototype, and the person must be told so, repeatedly
- The result you want is a surprise – confirmation teaches you nothing new

**KEY POINT** If the test confirmed everything you expected, you tested your own opinion and called it research.

# Thirty minutes with one user

## Thirty minutes with one user



*the twelve silent minutes in the middle are the whole point of the session*

Most of the value sits in the twelve minutes when you keep quiet.

# How many users you need

- Nielsen and Landauer's model, 1993: five users find roughly 85% of usability problems
- The first user finds about a third of them; each extra user finds fewer that are new
- So run three tests of five users rather than one test of fifteen
- Fix what you found between rounds – otherwise round two finds the same problems

**KEY POINT** Five users, fix, five more. Testing more people at once tells you the same thing more expensively.

# Writing the task

- Give a goal, not instructions: “get a drink of water before your next class”
- Never name the feature you want them to use
- Make it realistic enough that they bring their own habits to it
- Write down beforehand what you expect them to do, so you can be wrong on the record

**ACTIVITY** Ten minutes. Write one task for your prototype and your prediction of what the user will do. Seal the prediction until after the test.

# What to say, what not to say

## Say

- What are you thinking?
- What did you expect there?
- Tell me what you are looking for
- ... and then stay quiet

## Do not say

- No, click the other one
- It is supposed to do that
- Do you like it?
- Let me explain how it works

**KEY POINT** Every time you rescue the user, you delete the finding you came for.

# 2

## Capturing what happened

Feedback the whole team can act on, not one person's memory

# The feedback capture grid



## LIKES

what worked, in  
their words



## CRITICISMS

what frustrated  
or blocked them



## QUESTIONS

what they did not  
understand



## IDEAS

what they wished  
it did instead

Four quadrants, one note per observation, filled during the session.

# Filling the grid

- One observation per note, written while it happens, not afterwards
- Quote the user where you can; a paraphrase loses the evidence
- Put criticisms and questions in their own quadrants – a question is not a complaint
- After the session, cluster across users: one person is an anecdote, three are a pattern

**KEY POINT** The grid exists so that a disagreement about what to change becomes a disagreement about notes on a wall.

# A/B comparison

- Show two versions and ask the user to do the same task with each
- People are far better at comparing two things than at judging one
- Alternate which one comes first, or the second always looks better
- Ask which and why – the why is the finding, the which is just a vote

**KEY POINT** “Is this good?” gets you politeness. “Which of these two, and why?” gets you a reason.

# Three ways to test

| Method                  | What it answers             | When to use it              |
|-------------------------|-----------------------------|-----------------------------|
| <b>Task observation</b> | Can they actually do it?    | Any prototype, any fidelity |
| <b>A/B comparison</b>   | Which version works better? | Two options, one question   |
| <b>Field trial</b>      | Does it survive a real day? | Late, on a working model    |

# In class: test on five people

- Take your prototype outside the classroom and find five people who are not on your team
- One person runs the session, one fills the grid; swap after two users
- Fix the biggest problem between user three and user four, and note what changed
- Bring the filled grid to the next session

**ACTIVITY** Across the week. Five users, one grid, and one sentence on what you changed halfway through.

## THINK ABOUT IT

**Your five users all did something you did not predict.  
What does that tell you?**

- Was your prediction wrong, or your task?
- Did you intervene, and what did that cost you?
- Which finding would change the design most?
- Which finding are you most tempted to ignore?

# Summary

- Testing finds what people do; liking is easy to get and means nothing
- Thirty minutes: welcome, warm up, twelve silent minutes, debrief, close
- Five users find about 85% of problems – then fix and test five more
- Give a goal, never instructions, and record your prediction first
- Say almost nothing; every rescue deletes a finding
- Capture likes, criticisms, questions and ideas as you go, one note each

# Next: Iterative design and improvement

- Build, measure, learn – one question per loop
- Failing fast, and what that actually means
- Turning feedback into a decision
- When to stop iterating